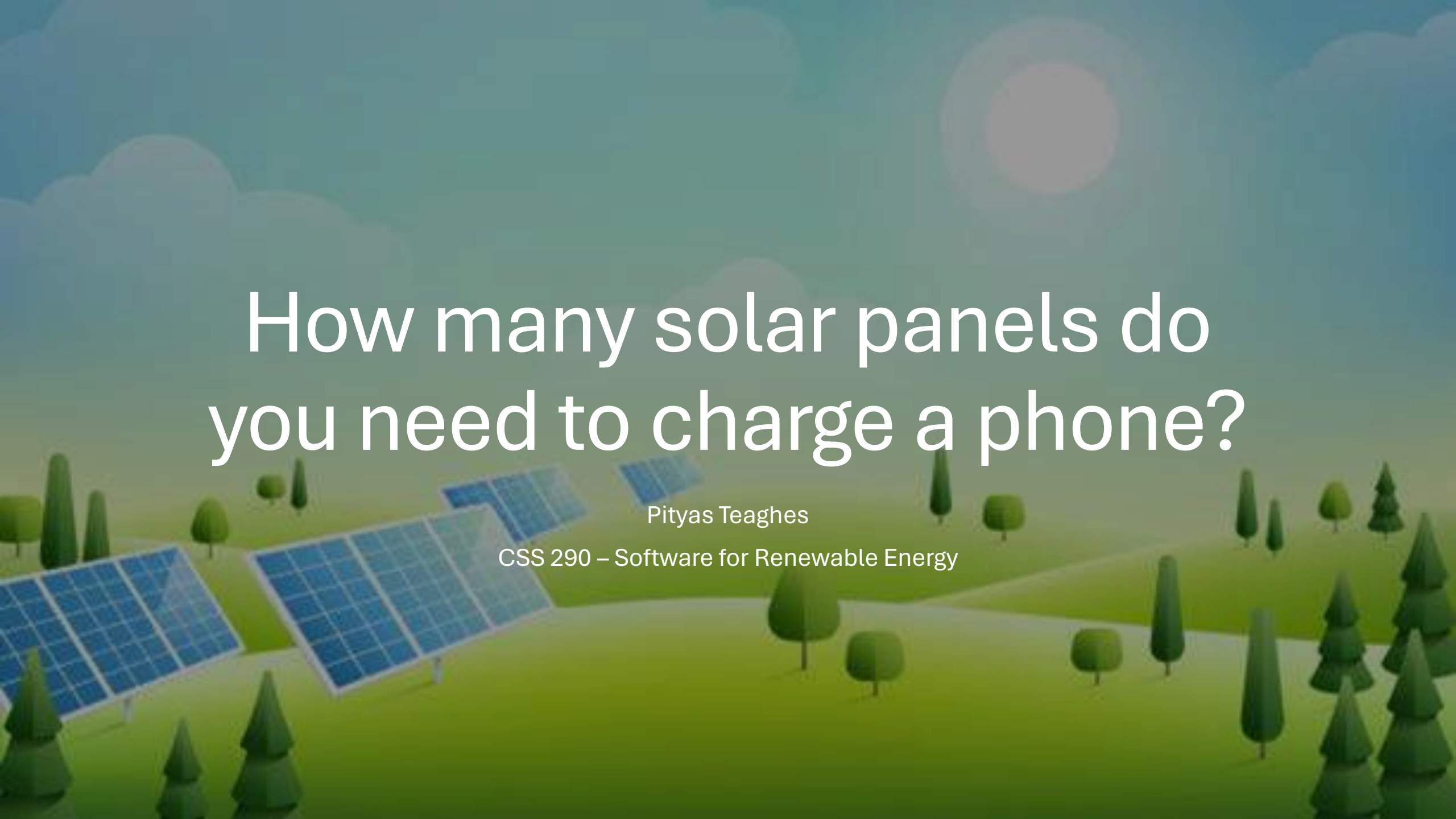


How many solar panels do you need to charge a phone?

Pityas Teaghes

CSS 290 – Software for Renewable Energy

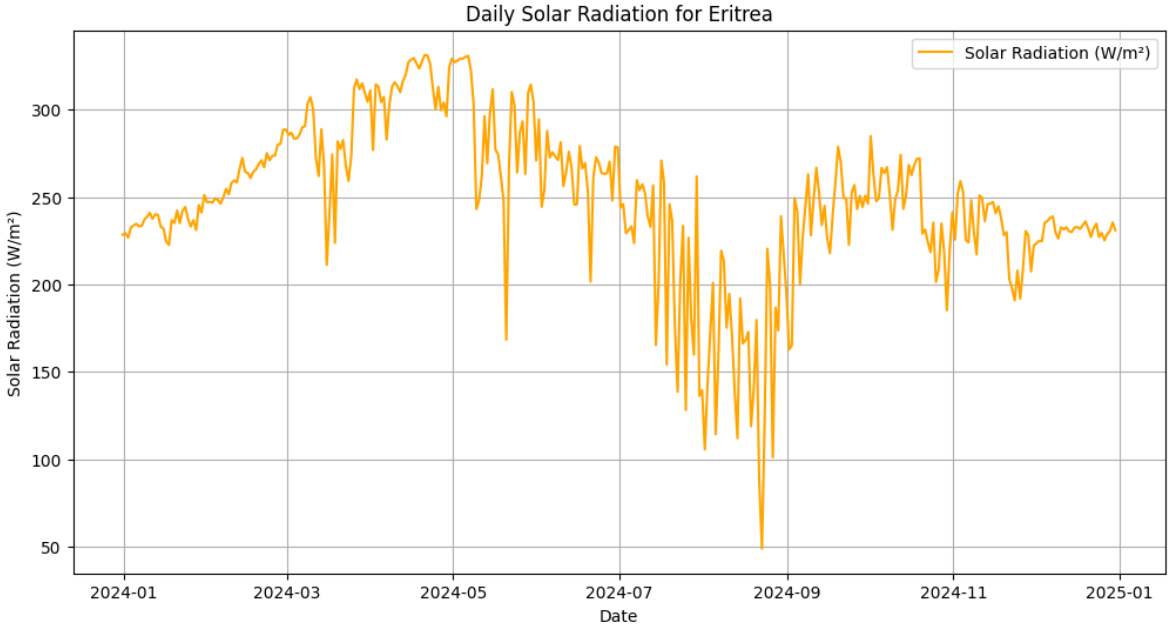
It depends on what you're charging

- **More Capacity = More Power Needed:** Larger batteries store more energy, so they require more solar panels to reach 100% in our simulation.

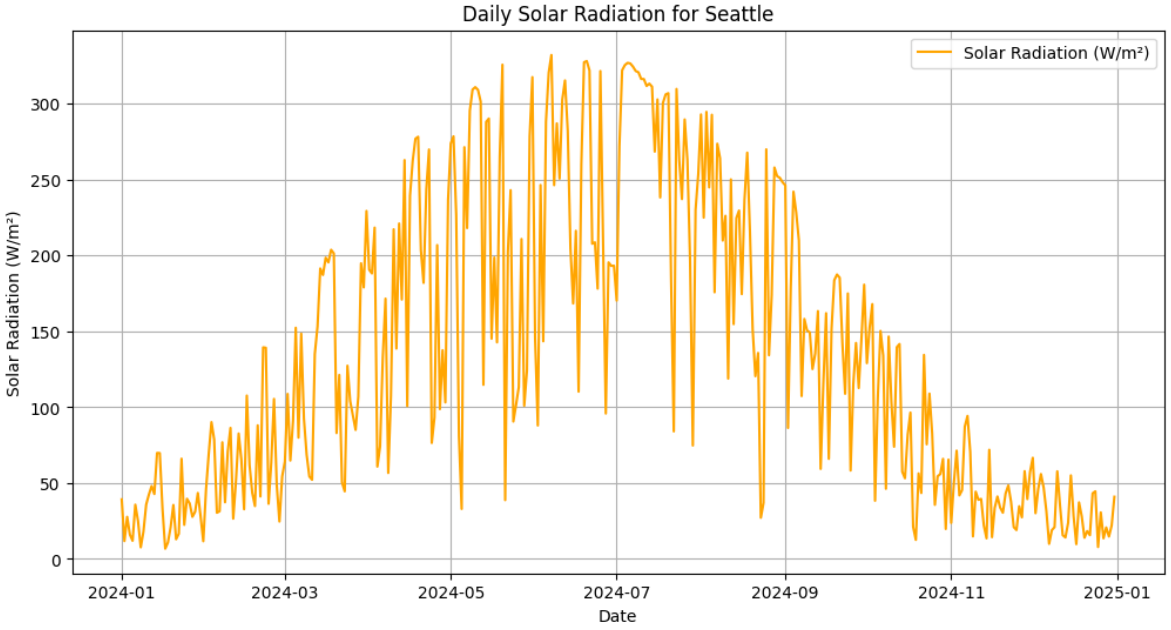
- **S25 Ultra vs. iPhone 15 Pro Max:** The S25 Ultra holds **500 mAh more** than the iPhone. This means it takes more solar energy and more time to fully charge than the iPhone.

Model	Capacity
iPhone 15 Pro Max	4422 mAh
Nothing Phone (2)	4700 mAh
Huawei P60 Pro	4815 mAh
Xiaomi 14 Pro	4880 mAh
Samsung Galaxy S25 Ultra	5000 mAh

Eritrea



Seattle



And it depends on where you live

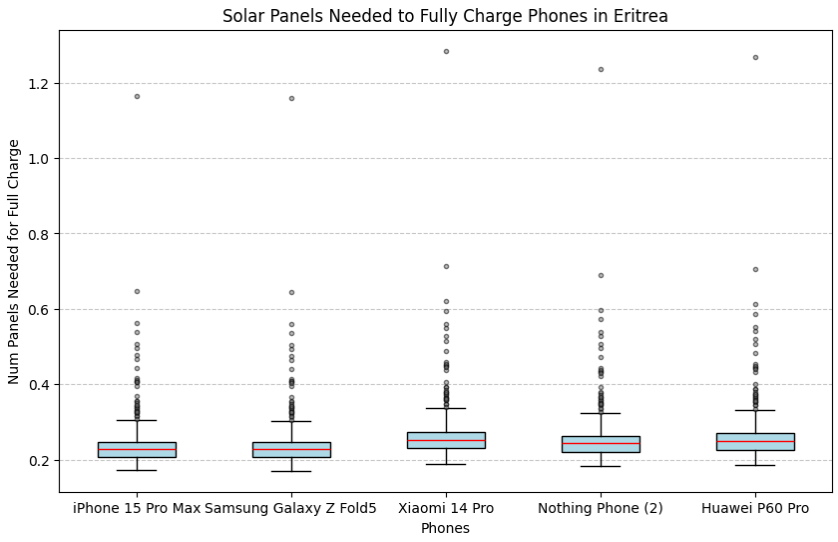
Visualization

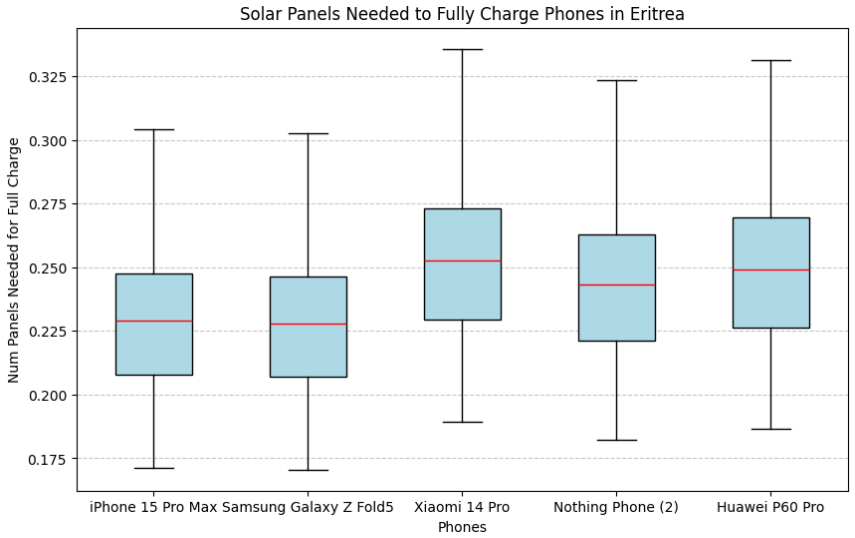
Why Box and Whisker Plots?

- Better Data Depth: While bar graphs only show the "top" and "bottom" of your data, box plots reveal the **Median**, **Quartiles**, and overall **Range** in one view.

- Clearer Distribution: They don't just show how much data you have—they show how it is **spread out** and if it is **skewed** toward one side.

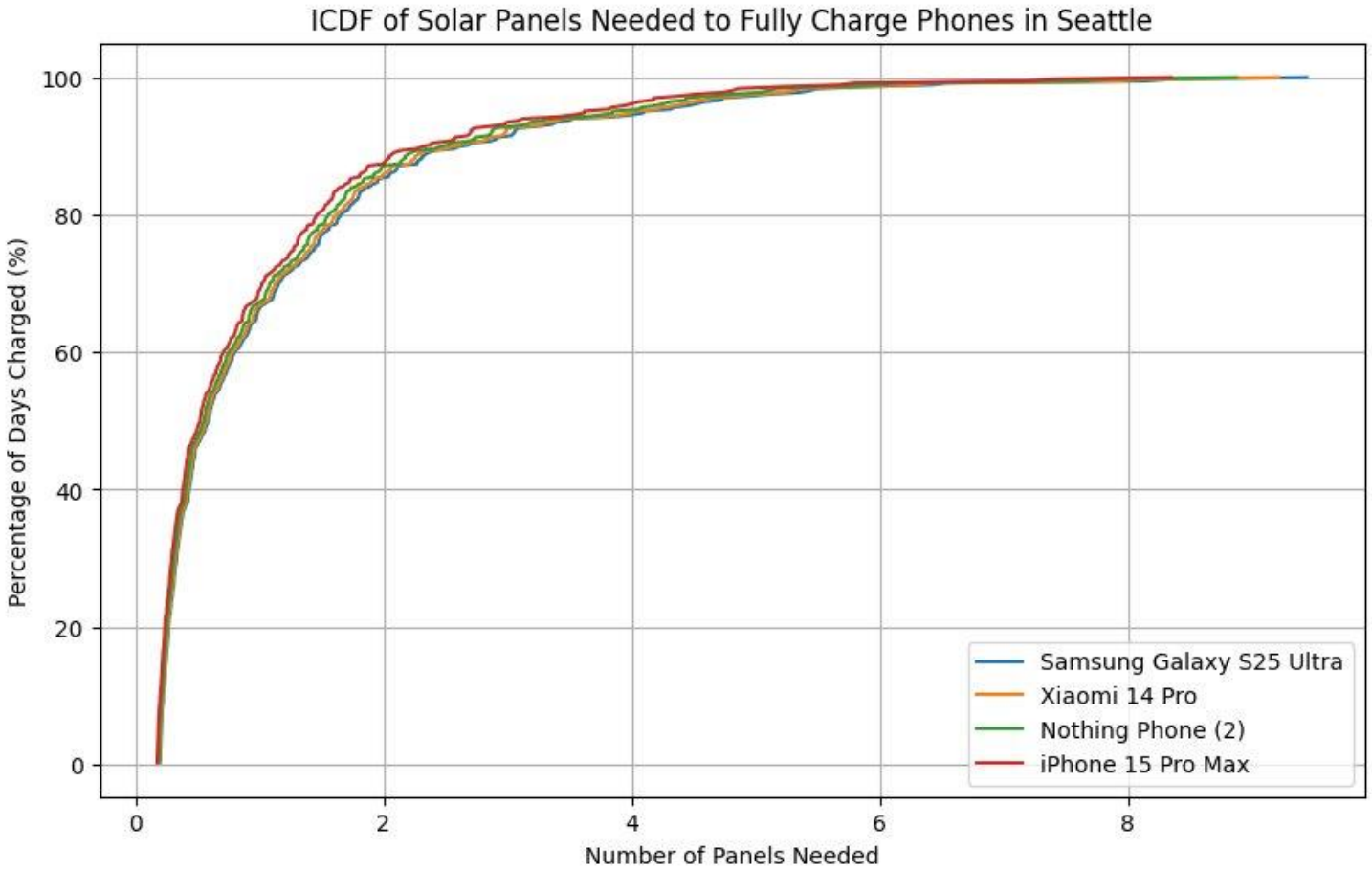
- Visual Simplicity: By summarizing thousands of data points into a single **"box,"** they remove the visual noise of individual dots, making the final output much easier to read.





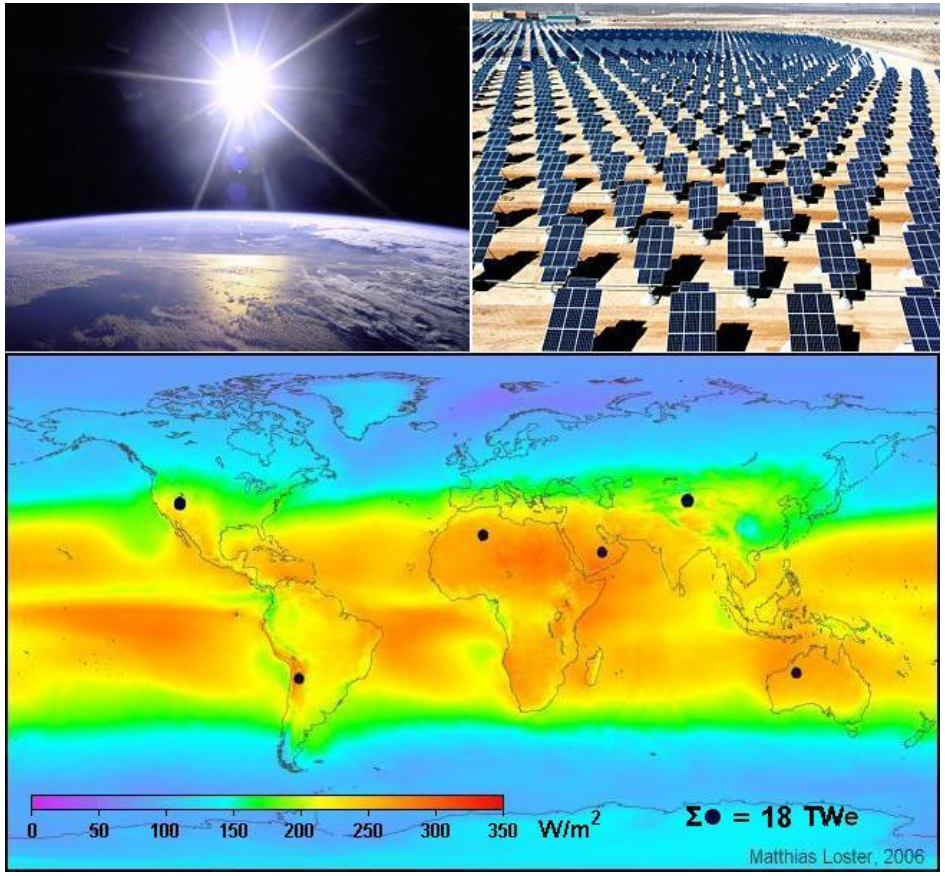
A different Visualization

If you look at this graph, you'll see that 90% of the year, you need only 2 panels to completely charge your phone.



Data Sources

- **Visual Crossings helped grasp a Weather API for solar panel performance estimation**
  - *It delivers critical data: solar energy, temperature, and cloud cover.*

- **Successfully used this API before.**
  - *Previously, it helped predict solar farm output in Program 3.*

- **The API offers key advantages for the project.**
  - It's simple, accurate, and directly supports estimating solar charging needs by location year-round.